BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 027 · Boolean algebra rules

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Normalise 0.0110×2^4 using a two’s-complement mantissa. Give the new mantissa and exponent.

Q2. A sorted array is [3, 8, 12, 19, 27, 35, 42] . When searching for 27 , which indexes are checked, in order?

Q3. Show the array after one full left-to-right pass of ascending bubble sort on [7, 3, 5, 2] .

Q4. An alarm sounds if the door is open AND the system is armed, OR if panic is pressed. Write Q using A, B and C.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

Recall the term — De Morgan

Recall the term — distribution

Recall the term — association

Recall the term — commutation

Recall the term — double negation

Recall the term — simplify

Recall the term — equivalent

New learning

De Morgan — CODE:
¬(A ∧ B) → ¬A ∨ ¬B
¬(A ∨ B) → ¬A ∧ ¬B — The operator inside the brackets swaps as each term is negated.

Double negation — CODE:
¬(¬B) → B — Two consecutive NOT operations cancel.

Distributive law — CODE:
(A ∧ B) ∨ (A ∧ C)
= A ∧ (B ∨ C) — A appears in both groups. Write it once outside the brackets, then join B and C with OR.

Association + commutation — CODE:
A ∨ (B ∨ C)
= (A ∨ B) ∨ C

A ∧ B
= B ∧ A — These allow regrouping and reordering before other simplification.

Worked example

Simplify ¬(A ∧ B)

1. Outer NOT over AND — CODE:
Use De Morgan's Law.

2. Negate each term — CODE:
A → ¬A
B → ¬B

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

3. AND becomes OR — CODE:
¬A ∨ ¬B

4. Equivalent behaviour — CODE:
Both are false only when A=1 and B=1.

We Do · Practise with guidance

De Morgan

Simplify ¬A ∨ ¬B using the relevant De Morgan form.

Complete the same expression

```
¬A ∨ ¬B
= ¬(? ? ?)
```

```
¬A ∨ ¬B
= ¬(A ∧ B)
```

Double negation

Simplify ¬(¬B).

```
¬(¬B)
= ?
```

```
¬(¬B)
= B
```

Distribution

Simplify (A ∨ B) ∧ (A ∨ C).

```
(A ∨ B) ∧ (A ∨ C)
= A ∨ (? ? ?)
```

```
(A ∨ B) ∧ (A ∨ C)
= A ∨ (B ∧ C)
```

Commutation + association

Rewrite C ∨ (A ∨ B) into (A ∨ B) ∨ C.

```
C ∨ (A ∨ B)
= ?
```

```
C ∨ (A ∨ B)
= (A ∨ B) ∨ C
```

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Simplify Boolean expressions
State the simplified form of each expression:
(i) ¬¬A (ii) ¬A ∧ ¬B (iii) ¬(¬A ∧ ¬B)

Question 2

Simplify the popcorn expression
A cinema gives free popcorn when a voucher is shown and either a soft drink or a bottle of water is purchased.
E = voucher shown; F = soft drink bought; G = water bought.
Simplify (E ∧ F) ∨ (E ∧ G) .

Question 3

Cinema discount: Boolean expression
A discounted cinema ticket is issued if the customer is under 18 AND has a student card , OR is over 60 AND has proof of age .
A = under 18; B = student card; C = over 60; D = ID. Write the Boolean expression for the output Q (discount ticket).

Question 4

From a gate circuit to an expression
Two OR gates take inputs A, B and C, D respectively. Their outputs feed a third OR gate. Write the complete Boolean expression for the circuit output.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Exit Ticket · Answer before checking

De Morgan's Laws swap AND and OR while negating each term.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. 0.1100×2^3 . Shift the mantissa one place left and decrease the exponent by one to preserve the value. A normalised positive two’s-complement mantissa begins 01 ; a negative one begins 10 .

Q2. Indexes 3 → 5 → 4 (values 19 → 35 → 27). Binary search tests the middle index of the remaining sorted range, using zero-based indexing.

Q3. [3, 5, 2, 7] . Compare adjacent values and swap when the left value is larger. The largest value, 7, moves to the end.

Q4. Q = (A ∧ B) ∨ C . A = door open, B = armed, C = panic pressed. Group door AND armed first; panic is an alternative trigger.

You Do pages · Independent questions

Q1. (i) Double negation
CODE:
¬¬A = A
Two NOT operations cancel.
(ii) De Morgan’s law
CODE:
¬A ∧ ¬B = ¬(A ∨ B)
NOT A AND NOT B means neither A nor B is true. This is an equivalent expression, not necessarily a shorter one.
(iii) De Morgan’s law, then double negation
CODE:
¬(¬A ∧ ¬B)
= ¬¬A ∨ ¬¬B
= A ∨ B

Q2. Step 1 — identify the repeated factor E.
CODE:
(E ∧ F) ∨ (E ∧ G)
Step 2 — use the distributive law.
CODE:
= E ∧ (F ∨ G)

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Read it back: the voucher must be shown AND either a soft drink OR water must be purchased.

Q3. Step 1 — under-18 route: under 18 AND student card.
CODE:
A ∧ B
Step 2 — over-60 route: over 60 AND ID.
CODE:
C ∧ D
Step 3 — either route qualifies, so combine with OR.
CODE:
Q = (A ∧ B) ∨ (C ∧ D)

Q4. Step 1 — first OR gate:
CODE:
X = A ∨ B
Step 2 — second OR gate:
CODE:
Y = C ∨ D
Step 3 — final OR gate joins X and Y:
CODE:
Q = X ∨ Y
= (A ∨ B) ∨ (C ∨ D)
Because OR is associative, this also equals A ∨ B ∨ C ∨ D.

Exit Ticket · Model answer

True. De Morgan's laws complement each input and exchange AND with OR when negating an expression.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Do Now Recall previous learning | **I Do** Read new knowledge | **We Do** Practise with guidance | **You Do** Work independently | **Review & Improve** Check understanding